

Chapter 4, Problem 98CP

Problem

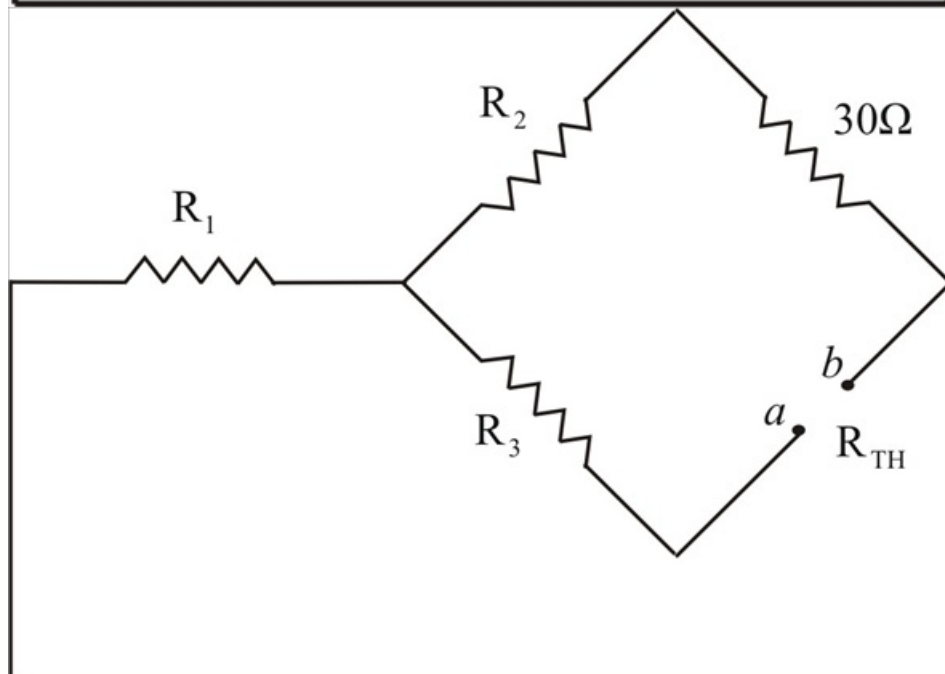
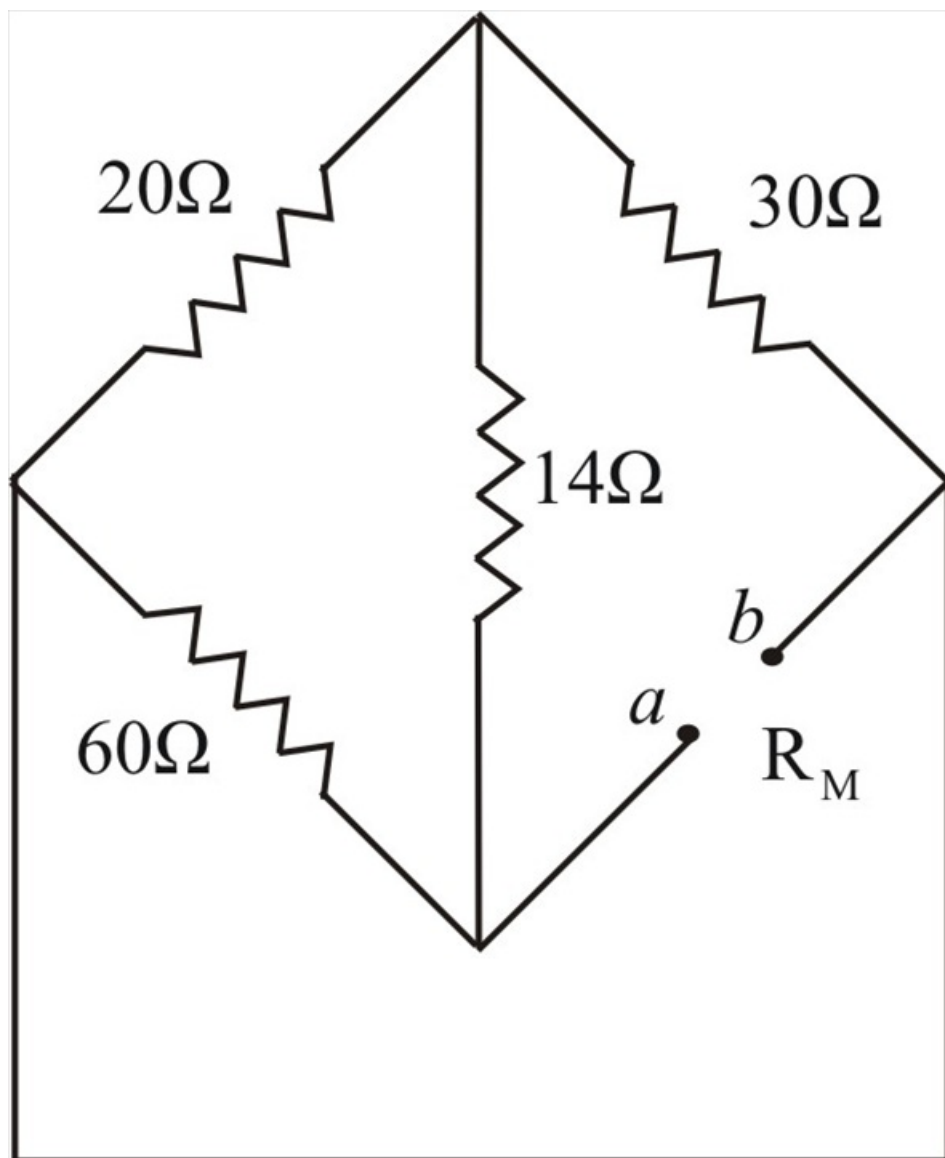
For Practice Prob. 4.18, determine the current through the $40\text{-}\Omega$ resistor and the power dissipated by the resistor.

Step-by-step solution

Step 1 of 5

The 20 -- ohm , 60 -- ohm , and 14 -- ohm resistors form a delta connection which needs to be connected to the WYE connection as shown in fig(b).

Step 2 of 5



$$R_1 = \frac{20 \times 60}{(20 + 60 + 14)} = \frac{1200}{94} = 12.766 \text{ Ohms} .$$

$$R_2 = 20 \times \frac{14}{94} = 2.979 \text{ Ohms} .$$

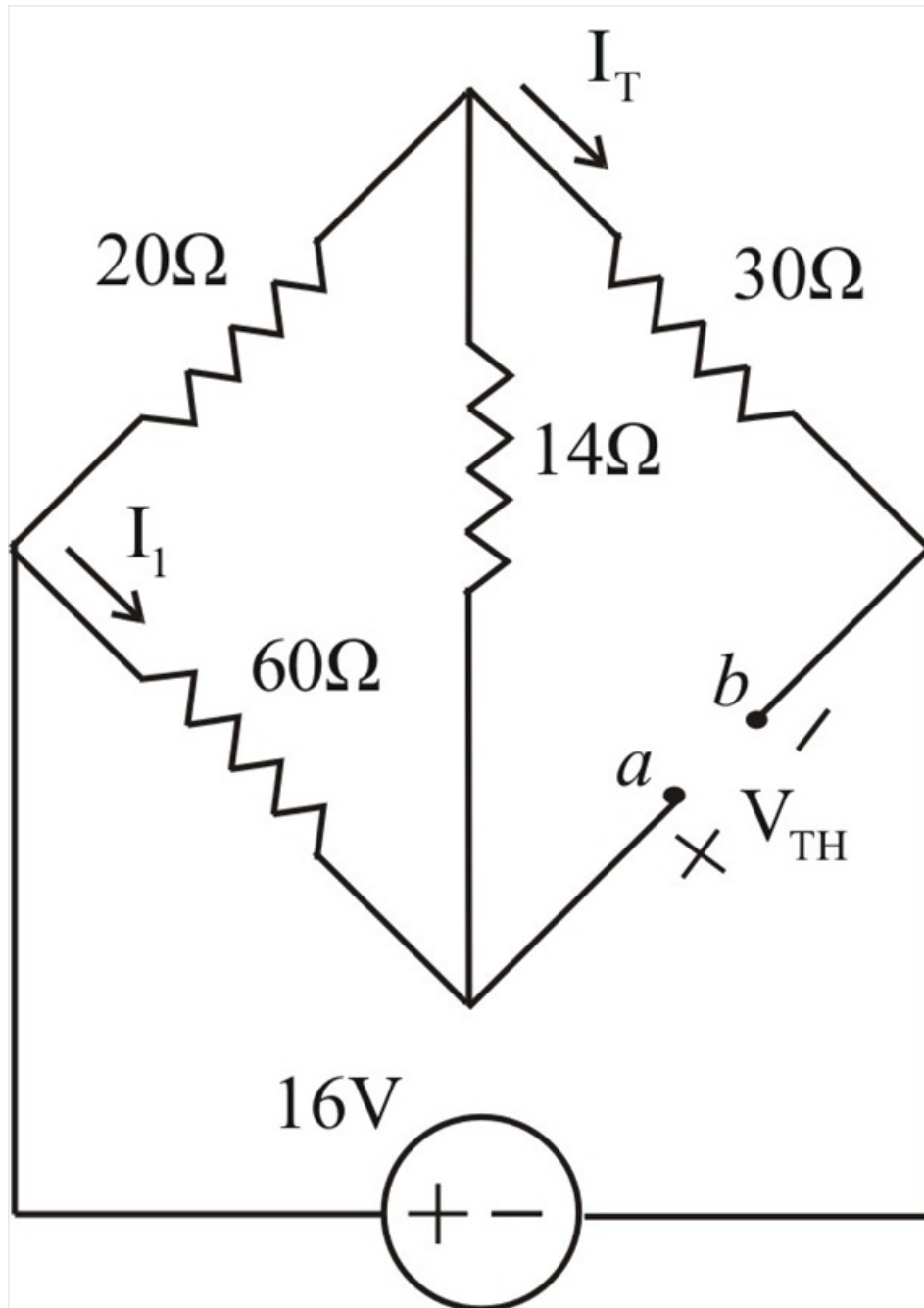
$$R_3 = 60 \times \frac{14}{94} = 8.936 \text{ Ohms} .$$

$$R_M = R_3 + R_1 \parallel (R_2 + 30) = 8.936 + 12.766 \parallel 32.98 .$$

= 18.139 Ohms.

To find V_{TH} consider the circuit in fig(c).

Step 4 of 5



Step 5 of 5

$$I_1 = \frac{16}{(30+15.745)} = 349.8 \text{ mA} .$$

$$I_1 = \left[\frac{20}{(20+60+14)} \right] I_T = 74.43 \text{ mA} .$$

$$V_{TH} = 14I_1 + 30I_T = 11.536 \text{ V}.$$

$$V_{40} = \frac{V_{TH}}{(R_{TH}+40)} = \frac{11.536}{(18.139+40)} = 198.42 \text{ mA} .$$

$$P_{40} = I_{40}^2 R = 1.5748 \text{ Watts} .$$